



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 2nd August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CE F434/BITS F494

Course Title : Environmental Impact Assessment

Instructor-in-charge : Dr. Pubali Mandal

1. **Course description:** Environment and global problems; Framing Environmental issues; effects of infrastructure development on environment; prediction and assessment of environmental impacts of infrastructure projects: technical and procedural aspects, guidelines and legal aspects of environmental protection, impacts on air, water, soil and noise environment, valuation, strategic assessment, mathematical modeling for environmental processes; social impact assessment (SIA), dislocation/disruption impact of Infrastructure projects; Life Cycle Assessments (LCA) and risk analysis methodologies; mitigation of environmental impacts; case studies; environmental management plan (EMP), national and international certification and guidelines including ISO.
2. **Scope and Objective:** The course focuses on understanding environmental issues linked to infrastructure development, emphasizing technical, procedural, and legal aspects of environmental impact assessment (EIA). It covers prediction and mitigation of impacts on air, water, soil, and noise environments, supported by modeling tools. Strategic assessments, LCA, risk analysis, and SIA are addressed alongside valuation and management frameworks. Case studies and EMP formulation are discussed with reference to national and international standards, including ISO, ensuring comprehensive exposure to certification protocols and regulatory compliance.
3. **Course learning outcomes:**
 - a. Identify and frame environmental issues arising from infrastructure development, including physical and social dimensions.
 - b. Apply procedural and technical approaches for impact prediction, assessment, and valuation using modeling tools and legal guidelines.
 - c. Formulate environmental management plans (EMP) and assess projects in line with national and international frameworks, including ISO.
4. **Text books and references:**

TB1. Y. Anjaneyulu, V. Manickam, Environmental Impact Assessment Methodologies, BS Publications, Third Reprint, 2013.

TB2. Larry W. Canter, Environmental Impact Assessment, McGraw-Hill, Second Edition, 1996.



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RB1. Kiely G., Environmental Engineering, Tata McGraw-Hill Indian Editions, 2007.

RB2. M. Anji Reddy, Environmental Impact Assessment: Theory and Practice, BS Publications, 2025.

5. **[Method of conduct of the course]** PowerPoint presentation and black board when necessary

6. Course Plan:

Week no.	Module	Lecture no.	Reference	Learning outcomes
1-2	Fundamental Approach to Environmental Impact Assessment	1-4	TB1	Understanding the basic concepts of EIA, what are the aim and objectives of EIA? and why it is carried out?
2	EIA Notifications (legal Aspects)	5-6	RB1	Idea about the legal aspects of EIA
3-5	EIA Methodologies	7-15	TB1, TB2	Understanding the different methodologies of EIA and which should be preferred for particular cases
6-7	Prediction and Assessments of impacts on soil and ground water environment	16-19	TB1, TB2	Impact prediction on soil and ground water, whether the impacts are positive or negative
7-10	Prediction and Assessments of impacts on surface water, biological environment, air environment, noise environment	20-28	TB1, TB2	Understanding the environmental attributes because of water , air and noise pollutants
11	Prediction and Assessments of impacts of socio – economic and human health aspects	29-31	TB1, TB2	Identification of social importance
12-13	Environment Management Plan (EMP), Disaster Management Plan (DMP)	32-35	Class Notes	Idea about the EMP and DMP and applications in Industries, Public facilities, Institutes etc.



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13-14	Environmental Auditing and Post Project Monitoring Case History	36-38	Class Notes, TB1	Understanding the real-life problems by field examples
14	ISO, Environmental Clearance, LCA	39-40	TB1, RB1	Importance and how to get the ISO certification for the industries, and formation of EC report

7. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quiz/Project	30%	40 min	TBA	Best 2 of 3, close book/open book
Mid-Semester	30%	90 min	TBA	Open/close book
Comprehensive	40%	180 min	TBA	Open/close book

Criterion for NC: If a student absents himself/herself in any component (listed in the Evaluation Scheme), his/her performance in the course may be reported as 'NC' (Not Cleared). A student should obtain 30% of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

8. Chamber consultation hours: Friday 5 PM to 6 PM, 1212 C

9. Makeup Policy:

Make-up will be granted only on genuine reasons.

For medical cases, a certificate from the Chief Medical Officer of the Medical Centre must be produced.

Make-up will not be conducted for class tests. However, best two out of three will be considered for final evaluation.

Pubali Mandal

Instructor-in-charge

[BITS F494]